

Software Requirements Specifications FOR AgroFlow: Agricultural Equipment Sharing Platform

Prepared by:

WANIA RAHMAN – GROUP LEADER (241830)

INSHA KHAN (241887)

ROMAN FATIMA (241878)

SANA IRAM (241884)

ANIQA IRSHAD (241866)

Submitted To: NAZISH YOUSAF

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1. Introduction

The **AgroFlow Agricultural Equipment Sharing Platform (Version 1.0)** is a scalable, service oriented digital platform enabling secure, transparent, and intelligent sharing of agricultural machinery across Pakistan.

This document defines the complete set of functional and technical specifications needed to direct system engineers, project sponsors, and selection panels throughout the design, construction, and deployment stages.

1.1 Purpose

The purpose of this document is to specify the software requirements for the AgroFlow Platform, Version 1.0. This document is intended for the project supervisor, the wider evaluation committee, system developers, and future maintainers. It provides a comprehensive breakdown of the system's features, user roles, and operational constraints to guide the development process.

1.2 Scope

AgroFlow is a distributed, service oriented digital platform designed to digitize and formalize Pakistan's agricultural equipment rental ecosystem. The system will serve farmers, equipment owners, and administrators across Pakistan's agricultural regions.

Core Capabilities (Phase 1):

- User management with role-based access control (farmers, owners, administrators)
- Digital asset registry for agricultural machinery with technical specifications
- Geographic search and filtering aligned with Pakistan's administrative hierarchy
- Manual booking engine with structured 10-state lifecycle management
- Logical escrow ledger for payment security and financial tracking
- Photo-based pre/post-rental inspection workflows
- Pakistan-specific KYC verification using CNIC and Fard documents
- Forensic activity logging with cryptographic hash chain for tamper detection

Extended Capabilities (Phase 2–3):

- IoT telematics integration for real time equipment tracking and engine diagnostics
- Predictive maintenance based on usage patterns and service history

- AI powered demand forecasting for supply demand optimization
- Dispute management workflow with structured escalation mechanisms
- Computer vision-based damage detection for objective condition assessment
- Smart route optimization using Dijkstra/A* algorithms
- Cooperative and group booking mechanisms for resource pooling
- Comprehensive revenue and tax reporting for regulatory compliance

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
AI/ML	Artificial Intelligence and Machine Learning for predictive analytics and automation
CNIC	Computerized National Identity Card, primary identity document in Pakistan
Fard	Official land record document establishing property ownership in Pakistan
KYC	Know Your Customer – identity and credential verification process
IoT Internet of Things –	networked devices collecting real-time equipment telemetry data
Telematics	Real-time tracking data including GPS location, engine hours, fuel consumption
Escrow Ledger	Logical accounting mechanism tracking payment states without fund custody
RBAC	Role-Based Access Control – permission system defining user capabilities by role
MQTT	Message Queuing Telemetry Transport – protocol for IoT device communication

QR Code	Machine-readable code enabling rapid equipment identification and tracking
Forensic Logging	Immutable audit trail recording all system actions with cryptographic verification
Hash Chain	Cryptographic sequence where each block includes hash of previous block
NADRA	National Database and Registration Authority – Pakistan's identity database
Term	Definition
FBR	Federal Board of Revenue – Pakistan's tax administration authority
PKR	Pakistani Rupee – Pakistan's official currency
SRS	Software Requirements Specification – detailed technical requirements document
BRD	Business Requirements Document – high-level business justification (this document)

1.4 References

- IEEE 830-1998 – Recommended Practice for Software Requirements Specifications
- AgroFlow Software Requirements Specification (Version 1.0, February 2025)
- Pakistan Electronic Transactions Ordinance 2002 – Electronic contracting framework
- Pakistan Personal Data Protection Bill (Draft) – Data privacy regulations
- OWASP Top 10 Security Standards – Web application security best practices
- Pakistan Agriculture Policy Framework 2021-2025 – National agricultural strategy
- Federal Board of Revenue Tax Reporting Guidelines – Digital platform compliance
- NADRA e-Sahulat Integration Guidelines – Identity verification integration
- State Bank of Pakistan Regulations on Digital Financial Services – Payment system rules
- Pakistan Bureau of Statistics Agricultural Census – Agricultural sector data

1.5 Overview

AgroFlow is a digital platform designed to transform agricultural equipment sharing in Pakistan by creating a trusted marketplace where farmers can access machinery on demand and equipment owners can maximize asset utilization. The platform addresses critical gaps in the current informal rental ecosystem through mandatory identity verification (CNIC/Fard), IoT-enabled asset tracking, forensic audit logging, and structured dispute resolution mechanisms. Built with an offline-first architecture suitable for rural connectivity constraints, AgroFlow implements logical escrow for payment security without requiring banking licenses. The system is designed for phased national deployment, with core marketplace functionality in Phase 1, followed by financial services and AI-powered features in subsequent phases, ultimately creating Pakistan's foundational digital infrastructure for agricultural mechanization.

2. Business Requirements

2.1 Background

Pakistan's Agricultural Landscape

Pakistan's agricultural sector is fundamental to the national economy, employing approximately 38% of the labor force and contributing nearly 23% to Gross Domestic Product (GDP). The sector provides raw materials to key industries and accounts for significant export earnings. However, Pakistani agriculture operates well below potential productivity due to structural inefficiencies, particularly in farm mechanization.

The Mechanization Challenge

The average farm size in Pakistan is less than 2.5 hectares, with smallholder farmers (owning less than 5 acres) constituting over 80% of all agricultural holdings. For these farmers, individual ownership of modern agricultural machinery is economically prohibitive, creating a fundamental barrier to mechanization.

Equipment Type	Average Cost (PKR)	% of Small Farmer Annual Income
Tractor (50-85 HP)	1,800,000 – 3,500,000	300–600%
Combine Harvester	4,500,000 – 8,000,000	750–1,400%
Wheat Thresher	350,000 – 600,000	60–100%
Rice Planter	250,000 – 450,000	40–75%

Cultivator/Rotavator	150,000 – 300,000	25–50%
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This capital intensity forces small farmers to rely on traditional labor-intensive methods that limit productivity and income.

Paradox of Underutilization

Paradoxically, equipment owners who can afford machinery face severe asset underutilization. Average annual tractor usage is 300–400 hours against potential capacity of 1,000–1,200 hours annually, representing utilization rates of only 30–35%. This underutilization represents significant wasted capital value of PKR 1.2–2.5 million per tractor annually.

The Informal Rental Ecosystem

In response to these challenges, an informal rental market has emerged, operating primarily through village level word-of-mouth networks, verbal agreements with no written terms, and cash transactions with minimal documentation. This informal ecosystem suffers from multiple structural problems affecting all stakeholders.

Pain Points for Farmers (Renters):

- Unreliable Access: No guaranteed equipment availability during critical operational windows
- Price Exploitation: Rates increase 100–200% during peak seasons without transparency
- Quality Uncertainty: No mechanism to verify equipment condition before rental
- Dispute Vulnerability: No recourse for breakdowns, poor performance, or unfair damage claims
- Documentation Gap: No rental history for credit applications or tax benefits

Pain Points for Equipment Owners:

- Trust Deficit: Reluctance to rent to unknown parties due to theft and misuse risks
- Payment Risk: Frequent delays, partial payments, or complete defaults with no recovery mechanism
- Asset Abuse: No visibility into how equipment is used, where it operates, or maintenance practices
- Damage Disputes: Difficulty proving pre-existing versus new damage
- Limited Market Reach: Unable to connect with renters beyond immediate social network

Systemic Economic Impacts:

- Productivity Loss: Delayed operations reduce crop yields by 15–25%
- Economic Informality: Transactions remain untaxed, unrecorded, invisible to policymakers

- Financing Barrier: No transaction history to enable equipment financing or farmer credit
- Regulatory Blindness: Government lacks data on mechanization levels and rental patterns

Technology Gap Analysis

Existing digital agriculture initiatives in Pakistan have failed to address the equipment rental space. Agriadvisory services focus on weather and crop advice; e-commerce platforms handle inputs only; fintech applications lack agricultural focus; and international platforms are designed for Western markets. No existing solution offers Pakistan specific KYC integration, physical asset tracking, forensic audit trails, or offline-first architecture required for rural deployment.

Rural Connectivity Constraints

Rural Pakistan faces significant connectivity constraints: 4G coverage reaches approximately 60% of agricultural areas; internet reliability is inconsistent; smartphone penetration remains limited in deep rural areas; and digital literacy varies across age groups. Any solution must address these constraints through appropriate architectural choices including offline-first synchronization and minimal data transfer requirements.

2.2 Business Opportunity

Problem Summary

The agricultural rental market in Pakistan operates almost entirely through informal contracts and trustbased transactions, creating a high-friction environment with significant risks for all parties including fraud, ownership disputes, pricing inconsistency, maintenance blindness, credit invisibility, and unresolved dispute escalation.

Market Gap

There is currently no centralized digital platform in Pakistan offering the following integrated capabilities:

Capability	Current Availability	AgroFlow Solution
Immutable Asset Provenance	None	Forensic logging with hash-chain verification
Escrow Secured Payments	Banking required	Logical escrow ledger without banking license

AI Demand Forecasting	None	Predictive availability and dynamic pricing
Offline First Architecture	None	Full synchronization with intermittent connectivity
Transaction-Based Credit Scoring	None	Trust tiers informed by verified rental history
Predictive Maintenance	Industrial only	Agricultural equipment focus
Pakistan KYC Integration	Partial	Full CNIC/Fard integration
Forensic Audit Logging	Enterprise systems	Accessible for all transactions

Market Size Estimation

Parameter	Estimate	Source/Basis
Total cultivated area	50 million acres	Pakistan Economic Survey
Operational tractors	500,000+	Agricultural Machinery Census
Annual tractor sales	50,000–70,000	Automotive Manufacturers Association
Average rental rate (tillage)	PKR 600–1,000/acre	Field surveys
Estimated annual rental market	PKR 30–50 billion	50M acres × PKR 800 × 75%

Annual growth potential	15–20%	Mechanization trend analysis
Target addressable market (Year 3)	5% = PKR 1.5–2.5 billion	Conservative penetration

Revenue Model

AgroFlow generates revenue through multiple sustainable streams:

Revenue Stream	Model	Margin Profile
Transaction Fees	1–3% of rental value	High volume, low per-transaction margin
Premium Listings	Monthly subscription for visibility	Recurring, high margin
Equipment Owner Subscriptions	Tiered access to analytics	Medium margin, recurring
Cooperative/Group Plans	Volume-based pricing	Medium margin, sticky customers
Data Insights	Aggregated analytics reports	High margin, specialized
Lead Generation	Referral fees to dealers	Commission-based
API Access	Third-party integration fees	Enterprise pricing

Competitive Positioning

AgroFlow differentiates through Pakistan specific identity integration leveraging existing NADRA infrastructure, physical asset tracking providing unavailable visibility, forensic audit trails creating legally defensible records, offline-first architecture designed for rural constraints, phased approach aligned with development capacity, logical escrow providing payment security without banking license, and contextual interface design for Pakistani agricultural practices.

2.3 Business Objectives

- **To digitize and formalize Pakistan's agricultural equipment rental ecosystem**, replacing informal verbal agreements with structured digital workflows.
- **To reduce capital ownership burden for small-scale farmers** through affordable, on-demand access to machinery via pay-per-use models.
- **To increase machinery utilization rates** from 300–400 to 800+ annual hours through efficient marketplace aggregation.
- **To implement escrow-based financial security** using logical escrow ledgers that provide payment protection without requiring banking licenses.
- **To establish blockchain backed asset authenticity** through immutable forensic logging for tamper-evident ownership and service histories.
- **To introduce AI-powered demand forecasting** enabling optimized supply planning and asset positioning.
- **To implement transaction-based credit scoring** for rental risk mitigation and verifiable trust tiers.
- **To enable predictive maintenance using IoT telemetry**, reducing breakdowns and extending equipment lifespan.
- **To ensure regulatory compliance** through automated reporting exports for FBR and government policy objectives.
- **To support scalable architecture** for national deployment from pilot districts to all provinces.
- **To integrate computer vision-based damage detection** for objective pre and post-rental condition assessment.
- **To enable cooperative and group booking mechanisms** allowing small farmers to pool resources.
- **To generate comprehensive agricultural mechanization data** for policymakers, researchers, and industry stakeholders.

2.4 Vision Statement

AgroFlow envisions becoming Pakistan's foundational digital infrastructure for agricultural mechanization, delivering a secure, intelligent, and scalable equipment sharing ecosystem.

The platform aspires to:

- **Modernize rural equipment access** through transparent, technology-enabled marketplaces.
- **Reduce operational risk** through identity verification, physical asset tracking, and structured dispute resolution.
- **Integrate AI and IoT into farming workflows**, making advanced technology accessible to small farmers.
- **Enable financial inclusion** through verified transaction histories supporting access to formal financing.
- **Provide legally defensible transaction records** via forensic audit trails meeting evidentiary standards.
- **Support national agricultural growth** through data intelligence on mechanization patterns.
- **Bridge the trust deficit** limiting equipment sharing, enabling confident renting and reliable access.
- **Democratize agricultural mechanization**, ensuring farm size does not determine access to productivity-enhancing technology.

2.5 Limitations and Exclusions

1. The system will NOT operate as a licensed banking institution, financial tracking uses logical escrow ledgers only, with no actual fund transfers.
2. The system will NOT manufacture or physically manage IoT hardware, it integrates with third party devices via APIs.
3. The system will NOT assume insurance or liability for equipment damage, only platform facilitated dispute mediation is provided.
4. The system will NOT integrate real time payment gateway processing in Version 1.0 — money movement is handled externally.
5. The system will NOT provide equipment financing or credit origination, credit scoring is for risk assessment only.